

Arjan Singh Puniani

Neural engineer · Neurotechnology founder · Researcher and educator
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EDUCATION

M.S. Bioengineering — University of Pittsburgh, Swanson School of Engineering (2023)

Graduate training in neural engineering, brain-computer interfaces, neural-signal analysis, medical devices, and quantitative systems neuroscience.

B.S. Chemical Engineering — University of California, Berkeley (2013)

Foundation in transport, thermodynamics, systems, computation, and quantitative engineering.

Post-baccalaureate science coursework — Clovis Community College (2015–2017)

Biology, chemistry, physics, anatomy, and supporting sciences.

SELECTED EXPERIENCE

Founder — Cyberpunk Reality (2022–Present)

Developing an early-stage focal cortical-cooling concept for drug-resistant epilepsy. Work includes thermal modeling, materials, prototype planning, and translational constraints. No claim of clinical validation or regulatory clearance.

R&D; Neural Engineer — Rehab Neural Engineering Labs, University of Pittsburgh (2022–2024)

Built psychophysical calibration tools and experimental interfaces; analyzed more than 940 calibration sessions; investigated longitudinal signal degradation in neural-interface data.

Neurophysiology Analyst — Sensorimotor Integration Laboratory (2021–2022)

Analyzed electrophysiology and eye-movement datasets using MATLAB and Hidden Markov Models.

Clinical Research Coordinator — UCSF Eureka Platform (2021)

Supported digital clinical-study operations, participant workflows, protocol development, and deployment readiness.

Chief of Staff — Rigetti Computing (2013–2015)

Supported early-stage operations, investor communications, and strategic execution.

MCAT Instructor — Altius Test Prep (2024–Present)

Teaches science reasoning and mentors medical-school applicants using concrete models for abstract concepts.

SELECTED PROJECTS

SeizeFreeze — Neurotechnology (Active prototype)

Focal cortical-cooling concept for drug-resistant epilepsy. Provisional intellectual-property wording and funding totals await primary-document confirmation.

Gamified BCI calibration — Neural engineering (Completed research work)

Experimental software and task design for demanding neural-perception calibration workflows.

Mechanism-to-Medical Center — Motorsport medicine (Under development)

Draft crash and concussion documentation toolkit circulated for expert review. It is not validated and is not a sanctioning-body standard.

ReasonOS Vector ECG — Software and education (Prototype)

Educational electrocardiogram measurement and reasoning interface with structured provenance. Not a diagnostic medical device.

TECHNICAL

Programming: Python, MATLAB, TypeScript, C++, R **Tools:** Git, Unity, SolidWorks, ANSYS, Blender **Domains:** neural interfaces, psychophysics, clinical research operations, medical-device translation